AI Network: An overlay network for consumers and providers in AI Ecosystem

Arashmid Akhavain arashmid.akhavain@huawei.com
Hesham Moussa hesham.moussa@huawei.com

Huawei Technologies Canada

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Background: AI Network Draft

- AI Systems are moving toward networked intelligence. What is an AI Network?
  - An overlay and layered network where models, data, compute resources, and agents interact.
  - It delivers a unified intelligent, multi-plane architecture for AI learning, inference, resource sharing, and agent interactions.
  - A platform that enables communications and services between providers and consumers in AI ecosystem.
  - Training, inference, and agent to agent collaboration/interaction can potentially have specific sub planes within the AI Network control, data, and OAM planes.

https://datatracker.ietf.org/doc/draft-akhavain-moussa-ai-network/01/

- The primary goal is to support dynamic and collaborative exchange of AI resources across a diverse ecosystem of stakeholders, including data providers, model developers, compute resource owners, and their consumers.

The proposed architecture serves to highlight areas that could benefit from standardised protocols and mechanisms, rather than being the focus of standardisation itself.

AI Network architecture, building blocks, and protocols

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AI service provider community

- Data providers
- Model providers
- Agents providers
- Resource providers
- Tools ???

AI client community

- Data clients
- Model clients
- Agent clients
- Resource clients
- Tools ???

AI Network Protocols CP, DP, OAM

AI Network Core

Others???

- X-RCE
- X-RRT
- S-FAM
- MDO
- TAG

AI Network Core Protocols CP, DP, OAM

- Tools Mgmt. ???
- Registration & Authentication
- Discovery & advertisement
- Billing/Accounting
- Trust & Reputation Mgmt.
- Upgrade Mgmt.
- Validation

AI Network Enablers

Communications/Networking

- Mobile network
- Internet
- NTN
- WiFi

Location

- GPS
- Mobile network
- Sensors

Others ???

AI Network architecture, building blocks, and protocols

AI Network service provider community

- Different providers (e.g. data, model, compute, agent owners) reside within this block.
- Providers join the network and become members via a registration and authentication process offered by the AI Network core.
- They use AI Network to advertise their services, capabilities, etc. across the network. They can register for notifications to get updates e.g. arrival of new models, training data, agents, etc.
- AI Network dispenses revenue to providers for the services rendered via its billing module.

AI Network client community

- This block consists of clients (e.g. other agents, models, etc.) requiring training, inference, agent to agent interactions, and those who need access to resources such as storage, compute, etc.
- Clients access services offered by members of the AI Network service provider community and/or AI Network core via different AI Network interfaces.

AI Network architecture, building blocks, and protocols

AI Network core

- Consists of network's main modules, functions, and services.
- Dedicated logical planes handle interactions between AI Network different core.
- modules and functions. Focus of AI Network Core Protocols.
- AI Network core interaction may or may not be visible/accessible to other blocks.

AI Network enablers

- This layer provides services that AI Network itself requires. Some examples are
  - A device/equipment hosting dataset, model, or an agent in mobile network might become idle over time. AI Network uses the underlay service to wake the hosting node before dispatching a job to it.
  - Tracking services offered by mobile networks or GPS.

AI Network protocols handle the interactions between AI Network core, communities, and enablers.

Inference

Modern AI inference is increasingly distributed and data-dependent.

- **Key drivers:**
  - High-volume query traffic from applications and agents
  - Distributed model deployments across cloud, edge, and devices
  - Cross-domain model access across providers
  - Retrieval-Augmented Generation (RAG) requiring real-time knowledge retrieval
  - Multi-model workflows and inference pipelines

- **Implications for networking:**
  - Low-latency data retrieval paths
  - Efficient model selection based on administrative domains and underlying network conditions
  - Coordination across compute, storage, and models

- **Inference therefore becomes a network-orchestrated process, not just model execution. It requires coordination across models, compute resources, and data sources across domains.**

Inference potential challenges

- **Advertisement and discovery service:**
  - Model providers need to discover resources for initial model deployment and subsequent instantiations of models if required. Resource providers of course need to be able to advertise their capabilities.
  - Models need to advertise their capabilities for query routing. Frontend entities (e.g. load balancers) need to be able to discover this information for scalability, reliability, and perhaps QoE.

- **Session management and reliability:**
  - Inference can be a lengthy process. Clients and model status need to be monitored as during this process either side might drop out (e.g. equipment failure, com link issues, etc.).

- **Privacy protection and trust establishment:**
  - Discovery information, advertisement, queries, and query responses may contain sensitive and private information that must be adequately protected.
  - The protocols might require to convey trust levels and ratings associated with models and queries.

- **Dataset discovery:**
  - Models providing inference services might need to discover new data to ensure uptodate responses (e.g. Retrieval Augmented Generation).

Inference in AI Network (proposed architecture)

- **AI Network orchestrate distributed inference across domains via the use of its CP, DP, and OAM planes**
  - Discovery of models via CP & descriptors
  - DP routes queries & streams results
  - Intelligence plane orchestrates multi-model flows
  - OAM tracks accuracy, latency, usage, and billing

- **Inference workflow in AI Network**
  - Client or agent publishes task descriptor
  - Models self-select or client selects via discovery
  - Intelligence plane sets up chaining/parallel graph
  - DP establishes session, streams inputs/outputs
  - Models return results; orchestration aggregates
  - OAM audits performance, enforces SLAs, bills usage

Concluding remarks

- **AI Networks ecosystem**
  - Infrastructure, data, models, services, orchestration, applications, platform, etc.

- **Identify common requirements**
  - Resource discovery and orchestration, trust, security, authorisation, registration, etc.

- **Separate the common from the specific**
  - Common cross-entity requirements → Address in broad, ecosystem-wide venues to enable interoperability and consistency.

  - Entity-specific work items → Address in specialized venues with sharper technical focus.

Align on the Common, Focus on the Specific

Thank You.

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